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### Tool changer for a robot and changing system therefor

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#### Abstract

The invention relates to a tool changer for a robot, including a base unit, a changing unit, which can be locked on the base unit, movable locking elements, which are provided on the base unit, receptacles, which are provided on the changing unit, the locking elements interacting, in a locking position, with the receptacles in order to lock the changing unit on the base unit and releasing, in an unlocking position, the receptacles, and a rotary element, which is provided on the base unit and is mounted for rotation about the longitudinal axis of the rotary element, the locking elements being movement-coupled to the rotary element in such a way that the locking elements are moved when the rotary element is rotated.

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## Field of Classification Search

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## Background/Summary

### CROSS REFERENCE TO RELATED APPLICATIONS

(1) This is a National Stage Application filed under 35 U.S.C. 371 based on International Patent Application No. PCT/EP2020/070214, filed on Jul. 16, 2020, which claims priority to German Patent Application 10 2019 119 657.4 filed Jul. 19, 2019. The entire disclosures of each of the foregoing are hereby incorporated herein by reference in their entireties.

### BACKGROUND

#### Field of Invention

(2) The invention relates to a tool changer for a robot, comprising a base unit, a changing unit

which can be locked on the base unit, movable locking elements which are provided on the base unit, and receptacles which are provided on the changing unit, the locking elements interacting with the receptacles in a locking position in order to lock the changing unit on the base unit and releasing the receptacles in an unlocking position, said tool changer also comprising a rotary element which is provided on the base unit and rotatably mounted about its longitudinal axis, the locking elements being movement-coupled to the rotary element in such a way that the locking elements are moved when the rotary element is rotated.

(3) The base unit is in particular designed in such a way that it can be arranged on the free end of a robot arm. The changing unit is in particular designed in such a way that it is configured for arrangement on a tool or gripping unit.

#### Description of Prior Art

(4) Robots are used in industrial assembly lines and for other manufacturing applications to perform repetitive tasks with a high degree of accuracy. For example, robots are commonly used in the automotive industry to perform a number of tasks such as material handling, cutting, welding, joining, and the like.

(5) A tool changer as mentioned at the outset is known, for example, from DE 112009002163 T5. The locking elements are designed as rolling bodies here. In this case, a piston with cam faces having several facets that act against the rolling bodies is known as a rotary element for activating the locking elements.

(6) Furthermore, a machine tool having a tool changing device is known from DE 33 46 467 A1. The tool changing device has a double-ended changing arm which is arranged so as to be pivotable about an axis parallel to the spindle axis and is provided with a tool gripper at each end.

(7) A tool changing apparatus for a welding robot is known from DE 40 10 070 C1. Two closure hooks are provided as locking elements. A piston acts on a toggle joint in a spring-loaded manner such that the closure hooks are spread apart and two base plates are oriented and braced against one another in a locking position. In order to release the connection, pressure is applied to the piston counter to the spring force.

(8) A gripper changing coupling is known from DE 37 13 619 A1. The locking mechanism comprises an adapter plate on which two centering pegs are provided. These are received in a centering bore and locked with a perpendicularly movable locking plate, the locking plate having two recesses, the wheels of which engage lockingly in undercuts on the centering pegs upon perpendicular movement.

#### SUMMARY OF THE INVENTION

(9) The problem addressed by the present invention is that of providing a tool changer of the type mentioned at the outset in which the locking elements act against the receptacles in the locking position with a relatively high degree of force. The locking elements should be moved as quickly as possible at the same time.

(10) This problem is solved by a tool changer having the features of claim 1. In a tool changer of this kind, the rotary element is in particular movement-coupled to a rotationally fixed and axially movable actuation element such that, when the rotary element is rotated, the actuation element changes its position in the axial direction. The rotary element as such can also change its axial position; however, it is also conceivable for the rotary element and the actuation element to be movement-coupled to one another in such a way that the rotary element does not change its axial position when it is rotated. Furthermore, the actuation element is movement-coupled to or forms a pivot pin which extends perpendicularly to the longitudinal axis of the actuation element, a joint pin is provided on each of the locking elements, and, between the pivot pin and each joint pin, at least one lever arm is provided. Each lever arm is designed in such a way that, when the pivot pin is moved in the axial direction, the joint pins and thus also the locking elements are moved in a perpendicular direction running perpendicularly to the axial direction. A plurality of lever arms can also be provided between the pivot pin and each joint pin. Providing such an activation mechanism

for the locking elements has the advantage that, by using the lever arms, the locking elements can cover relatively long distances with relatively little force as long as they are not in the locking position. Then, when the locking elements come into the region of the locking position, they cover relatively short distances while providing a relatively high degree of force. By providing the lever arms, the movement of the actuation element in the axial direction can also be advantageously deflected into a movement of the locking elements in the perpendicular direction. Overall, the movement of the pivot pin and the joint pin can also take place in relatively wear-free manner.

(11) It is advantageous for the joint pin and the pivot pin to be circular-cylindrical and arranged so as to run parallel to one another.

(12) In contrast to the known prior art as represented by DE 11 2009 002 163 T5, no rolling bodies which are exposed to high loads and high wear due to the relative movements of the components movement-coupled thereto are used.

(13) Furthermore, it is also advantageous for the rotary element to comprise a head and a threaded rod, the head being rotatable by means of a rotary tool and the threaded rod being rotatably mounted in a threaded sleeve that is arranged on the base unit in a rotationally fixed manner. The threaded rod and the threaded sleeve can interact directly; the thread of the threaded rod then engages in a thread of the threaded sleeve. However, it is also conceivable for rolling bodies to be provided between the threaded rod and the threaded sleeve in order to minimize the friction between the threaded rod and the threaded sleeve. The rotary tool can be a hand-operated rotary tool or a machine-operated rotary tool. Providing a machine-operated rotary tool has the advantage that the rotary element can be operated in an automated manner.

(14) The actuation element is advantageously provided on the rotary element on the side of the rotary element facing away from the head so as to be rotationally decoupled in the axial direction. The rotational decoupling can be implemented, for example, in such a way that the actuation element passes through and undercuts a coupling portion of the rotary element.

(15) In a further advantageous embodiment, the head has an engagement surface for the rotary tool and a securing element which interacts with the engagement surface to secure the rotary element against rotation is provided on the base unit. The securing element is designed in such a way that it disengages from a securing position into a release position when the rotary tool is placed on the head or when the rotary tool is rotated. The securing element can, for example, be a securing wedge which is spring-loaded in the axial direction and which abuts a flat portion of the head and thereby prevents the rotary element from rotating. When the rotary tool is attached, this securing wedge can be moved backward in the axial direction from the securing position into the release position, as a result of which the rotary element can be rotated.

(16) Furthermore, the two lever arms are preferably arranged one above the other and/or so as to adjoin one another, at least in portions. This results in a space-saving arrangement.

(17) It is also advantageous for the locking elements to be designed as slide elements mounted in the base unit in a guided manner. The slide elements are in particular mounted so as to be guided in such a way that they can only move in the direction of rotation. In this way, a clear direction of movement of the slide elements can be specified.

(18) The slide elements can in particular provide wedge portions on the side facing the respective receptacles. The receptacles can in particular be designed as complementary wedge receptacles.

(19) It is also advantageous for the receptacles to be designed to be at least partially elastically yielding and/or to be arranged on the changing unit so as to be at least partially elastic. The receptacles can thus be deflected in an elastically yielding manner when the locking position is reached, as a result of which the locking elements act against the receptacles under a preload in the locking position. The advantage of providing such a preload is that it is possible to compensate for play resulting in particular from manufacturing tolerances.

(20) It is advantageous if spring elements are provided which act on the receptacles against the changing unit in a spring-preloaded manner. In particular, spiral or disc springs can be used as

spring elements.

(21) It has also proven to be advantageous if, in the unlocking position, the connecting axes between the respective joint pins and the pivot pin form an angle  $\alpha$  with the axial direction in the range of from 30° to 60°, and preferably in the range of from 40° to 50°, and/or if, in the locking position, the connecting axes between the respective joint pins and the pivot pin form an angle  $\alpha$  with the axial direction in the range of from under 94° to over 90°. This ensures that a high degree of force transmission can be provided by the lever arms, particularly in the locking position; consequently, locking takes place with a relatively high degree of force overall.

(22) It is also advantageous for a displaceable axial stop to be provided for adjusting the end position of the actuation element in the axial direction. When the end position is adjusted, the angle  $\alpha$  can ultimately also be adjusted in the locking position.

(23) According to a further embodiment of the invention, it is conceivable for positioning recesses or positioning rods by means of which the changing unit can be precisely positioned at a changing station to be provided on the changing unit. This makes it possible to ensure that the changing unit, in particular when it is detached from the base unit, can be mounted at a changing station.

(24) The aforementioned problem is also solved by a changing system comprising a tool changer according to the invention, also comprising a changing station, wherein the changing station has a rotary tool for rotating the rotary element and wherein the changing station has positioning rods or positioning recesses by means of which the changing unit can be positioned at the changing station. A fully automated activation of the tool changer can therefore be achieved by such a changing system.

(25) Further details and advantageous embodiments of the invention can be found in the following description, on the basis of which one embodiment of the invention will be described and explained in more detail.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings:

(2) FIG. 1 shows a changing system comprising a tool changer and comprising a changing station;

(3) FIG. 2 shows the tool changer according to FIG. 1;

(4) FIG. 3 shows the changing system according to FIG. 2 with the changing unit unlocked;

(5) FIG. 4 is a section from the changing system according to FIG. 1 when moving the tool changer to the changing station;

(6) FIG. 5 is the view according to FIG. 4 with the tool changer arranged at the changing station;

(7) FIG. 6 is a longitudinal section through the changing system according to FIG. 3;

(8) FIG. 7 is a cross section through the changing system according to FIG. 8;

(9) FIG. 8 is a cross section corresponding to FIG. 7 in the locking position;

(10) FIG. 9 is a detail of a plan view of the base part without the housing;

(11) FIG. 10 shows the locking elements of the base unit with the associated movement mechanism in the unlocking position; and

(12) FIG. 11 is a view corresponding to FIG. 10 in the locking position.

### DETAILED DESCRIPTION

(13) A tool changer **10** which comprises a base unit **12** and a changing unit **14** is shown in the drawings. A changing station **16** at which the tool changer **10** can be docked in order to release the changing unit **14** from the base unit **12** is also shown, in particular in FIGS. 1 to 5. The tool changer **10** and the changing station together form a changing system **18**.

(14) The base unit **12** can be arranged on the free end of a robot arm. Using the robot arm, which is not shown in the drawings, the base unit **12** can be moved freely in space and, inter alia, can also be

moved toward the changing station **16**. During operation, in particular a tool or a handling system, such as a gripping device, is arranged on the changing unit **14** in order to process or handle components. The changing unit **14** can be detached from the base unit **12** in order to exchange the tool or the handling unit. For this purpose, the tool changer **10**, i.e. the base unit **12** together with the changing unit **14**, is moved to the changing station **16**, as is shown in FIG. 2. In order to achieve positionally accurate docking of the tool changer **10** at the changing station **16**, three positioning rods **20** are provided on the changing station **16**, which rods can enter the positioning recesses **22** provided on the changing unit **14**, as indicated in FIG. 6.

(15) As can be seen in FIGS. 1 to 5, a rotary element **24**, which is mounted so as to be rotatable about its longitudinal axis **25**, is provided on the base unit **12**. FIGS. 4 and 5 show the housing of the base unit **12** with dashed lines only in order to make the components within the base unit **12** visible.

(16) As can be seen in particular from the section according to FIG. 6, the rotary element **24** comprises a head **26** and a threaded rod **28**. The head **26** has engagement surfaces on two opposite sides in the form of flat portions **30** for a rotatably drivable rotary tool **32**. As can be seen from FIGS. 1 to 5 in particular, the rotary tool **32** has a fork-like design and is arranged on the changing station **16** so as to be rotatably drivable. The rotary tool **32** has engagement surfaces **34** which are formed parallel to the flat portions **30**. When the tool changer **10** is docked at the changing station **16**, the engagement surfaces **34** of the rotary tool **32** consequently abut the flat portions **30** of the head **26**, as can be seen from FIGS. 2 and 5.

(17) As can be seen in particular from FIGS. 4 and 5, a securing element **36** is also provided on the base unit **12**, which securing element interacts with one of the flat portions **30** to secure the rotary element **24** against rotation. The securing element **36** is preloaded by means of spring elements **38** in the axial direction, i.e. in the direction of the longitudinal axis **25** toward the changing station **16**. Rotation of the rotary element **24** is thus not possible when the tool changer **10** is not docked at the changing station **16**. If the tool changer **10** is located at the changing station **16**, as shown in FIG. 5, the rotary tool **32** interacts with the securing element **36** in such a way that, when the rotary tool **32** is rotated, it disengages from the securing position of the securing element **36** shown in FIG. 4 into a release position shown in FIG. 5. In order to achieve safe disengagement against the preloading force of the spring elements **38**, each of the mutually facing sides of the securing element **36** and of the rotary tool **32** is wedge-shaped.

(18) As can be seen in particular from the section according to FIG. 6, the threaded rod **28** of the rotary element **24** is rotatably mounted in a threaded sleeve **40**. In order to achieve the smoothest possible rotation of the rotary element **24** in the threaded sleeve **40**, rolling bodies are provided between the threaded rod **28** and the threaded sleeve **40**. However, it is also conceivable for the threaded rod **28** to act directly against the threaded sleeve **40**. By providing the threaded sleeve **40**, it becomes possible for the rotary element **24** to change its position in the axial direction, i.e. in the direction along the longitudinal axis **25**, when the rotary element **24** is rotated about its longitudinal axis **25**. The axial direction is indicated by the double arrow **41**.

(19) As can also be seen from FIGS. 4 to 8, the rotary element **24** is movement-coupled in the axial direction **41** to a rotationally fixed and axially movable actuation element **42**. The actuation element **42** adjoins the rotary element **24** on the side of the rotary element facing away from the head **26** so as to be rotationally decoupled in the axial direction **41**. For rotational decoupling and simultaneous axial movement coupling, the rotary element **24** has, as can be seen from FIG. 6, an undercut **44** which engages behind an opening **46** of the actuation element **42**, which opening faces the rotary element **24**. This ensures that rotation of the rotary element **24** results in movement of the actuation element in the axial direction **41** without the actuation element **42** being rotated about the longitudinal axis **25**.

(20) As can also be seen in particular from FIG. 6, the actuation element **42** has a bore **48** which extends perpendicularly to the longitudinal axis **25** and in which a pivot pin **50** is provided. The

pivot pin **50** consequently extends perpendicularly to the longitudinal axis **25** or perpendicularly to the axial direction **41**.

(21) As can also be seen from FIG. **6**, the pivot pin **50** extends not only through the bore **48** of the actuation element **42**, but also through two further bores **52**, which are provided on respective lever arms **54**, **56**. The two lever arms **54** and **56**, which can be seen in particular in FIGS. **9** to **11**, each interact with respective joint pins **58**, which are in turn provided on respective locking elements **60**, **62**.

(22) Receptacles **80** for locking the changing unit **14** to the base unit **12** are shown on the changing unit in FIGS. **3**, **6**, and **7**, which receptacles are designed as wedge grooves **84** provided in bridge portions **82**. The locking elements **60**, **62** that interact with the receptacles **80** are provided on the base unit **12** and each have a wedge portion **86**, which is designed to be complementary to the wedge grooves **84**, at the free ends thereof. The locking elements **60**, **62** having the wedge portions **86** can be seen in particular in FIGS. **9** and **10**.

(23) By moving the actuation element **42** in the axial direction, as illustrated in FIG. **10**, the two locking elements **60**, **62** can be moved from an unlocking position shown in FIG. **10** into a locking position shown in FIG. **11**. For this purpose, the locking elements **60**, **62** move along a perpendicular axis **64** that runs perpendicularly to the longitudinal axis **25** in a perpendicular direction indicated by the arrows **68**. If, according to FIG. **10**, the actuation element **42** is moved in the direction of the arrow **66** along the longitudinal axis **25** in the axial direction **41**, the locking elements **60**, **62** are moved along the perpendicular axis **64** in the perpendicular direction **68** due to the mechanism of the lever arms **54**, **56**.

(24) In the unlocking position of the locking elements **60**, **62** shown in FIG. **10**, the connecting axes **70**, which run between the centers of the joint pins **58** and the center of the pivot pin **50**, each form an angle  $\alpha$  of  $45^\circ$  with the longitudinal axis **25**. In the locking position shown in FIG. **11**, the axes **70** form an angle  $\beta$  of approximately  $88^\circ$  with the longitudinal axis **25**. This means that, proceeding from FIG. **10**, the locking elements **60**, **62** are first moved relatively quickly in the direction of the arrows **68** with relatively little force. When the locking position, as shown in FIG. **11**, is reached, the movement of the locking elements **60**, **62** is slowed down due to the kinematics provided, but the locking force is increased accordingly. The angle  $\beta$  is selected as to be less than  $90^\circ$ .

(25) In order to ensure that the angle  $\beta$  is less than  $90^\circ$ , an axial stop **72** that is displaceable in the axial direction **41** is shown in FIG. **11**. The axial stop **72** is designed here as a threaded pin which can be screwed into the housing of the base unit **12**, as shown in FIG. **9**.

(26) In order to ensure that the locking elements **60**, **62** move along the perpendicular axis **64** when the actuation element **42** is moved, guide pockets **74**, shown in FIG. **9**, in which the locking elements **60**, **62** are slidably arranged are provided in the housing of the changing unit **14**.

(27) If, proceeding from FIG. **11**, the actuation element **42** is moved back, i.e. to the right, by rotation of the rotary element **24** in the other direction, the locking elements **60**, **62** move from the locking position shown in FIG. **11** back into the unlocking position shown in FIG. **10** along the perpendicular axis **64** counter to the directions of the arrows **68** in FIG. **10**.

(28) In order to ensure a sufficient preloading force in the locking position, the bridge portions **82** together with the receptacles **80** are arranged on the changing unit **14** in a partially elastically yielding manner. In particular, the bridge portions **82** can be provided on the changing unit **14** such that they can be deflected by means of spring elements. If, as shown in FIG. **8**, the locking elements **60**, **62** are acted on along the arrows **68** against the receptacles **80** in the locking position, said receptacles are deflected into the deflection position **82'** shown by dashed lines in FIG. **8**. The advantage of this is that it is possible to reliably compensate for play that occurs, for example, due to tolerances. The deflection of the bridge portions **82** or the receptacles **80** is greater than the overall tolerance.

## Claims

1. Tool changer for a robot, comprising a base unit, a changing unit which can be locked on the base unit, movable locking elements which are provided on the base unit, and receptacles which are provided on the changing unit, the locking elements interacting with the receptacles in a locking position in order to lock the changing unit on the base unit and releasing the receptacles in an unlocking position, said tool changer also comprising a rotary element which is provided on the base unit and is rotatably mounted about its longitudinal axis, the locking elements being movement-coupled to the rotary element in such a way that the locking elements are moved when the rotary element is rotated, characterized in that the rotary element is movement-coupled to a rotationally fixed and axially movable actuation element such that, when the rotary element is rotated, the actuation element changes its position in the axial direction, in that the actuation element is movement-coupled to or forms a pivot pin which extends perpendicularly to the longitudinal axis of the actuation element, in that a joint pin is provided on each of the locking elements, and in that, between the pivot pin and each joint pin, at least one lever arm is provided such that, when the pivot pin is moved in the axial direction, the joint pins and thus also the locking elements are moved in a perpendicular direction running perpendicularly to the axial direction, wherein the rotary element comprises a head and a threaded rod, the head being rotatable by means of a rotary tool and the threaded rod being rotatably mounted in a threaded sleeve that is arranged on the base unit in a rotationally fixed manner.
2. Tool changer according to claim 1, characterized in that the actuation element adjoins the rotary element on the side of the rotary element facing away from the head so as to be rotationally decoupled in the axial direction.
3. Tool changer according to claim 1, characterized in that the head has an engagement surface for the rotary tool and in that a securing element which interacts with the engagement surface in order to prevent rotation of the rotary element is provided on the base unit, the securing element being designed in such a way that it disengages from a securing position into a release position when the rotary tool is placed on the head or when the rotary tool is rotated.
4. Tool changer according to claim 1, characterized in that the at least one lever arm comprises two lever arms that are arranged one above the other and so as to adjoin one another, at least in portions.
5. Tool changer according to claim 1, characterized in that the locking elements are designed as slide elements mounted so as to be guided in pockets of the base unit.
6. Tool changer according to claim 5, characterized in that the slide elements have wedge portions on the side facing the respective receptacles and in that the receptacles are designed as wedge receptacles.
7. Tool changer according to claim 1, characterized in that the receptacles are designed to be at least partially elastically yielding when the locking position is reached.
8. Tool changer according to claim 7, characterized in that the receptacles are arranged on the changing unit so as to be at least partially elastically yielding when the locking position is reached.
9. Tool changer for a robot, comprising a base unit, a changing unit which can be locked on the base unit, movable locking elements which are provided on the base unit, and receptacles which are provided on the changing unit, the locking elements interacting with the receptacles in a locking position in order to lock the changing unit on the base unit and releasing the receptacles in an unlocking position, said tool changer also comprising a rotary element which is provided on the base unit and is rotatably mounted about its longitudinal axis, the locking elements being movement-coupled to the rotary element in such a way that the locking elements are moved when the rotary element is rotated, characterized in that the rotary element is movement-coupled to a rotationally fixed and axially movable actuation element such that, when the rotary element is

rotated, the actuation element changes its position in the axial direction, in that the actuation element is movement-coupled to or forms a pivot pin which extends perpendicularly to the longitudinal axis of the actuation element, in that a joint pin is provided on each of the locking elements, and in that, between the pivot pin and each joint pin, at least one lever arm is provided such that, when the pivot pin is moved in the axial direction, the joint pins and thus also the locking elements are moved in a perpendicular direction running perpendicularly to the axial direction, characterized in that, in the unlocking position, the connecting axes between the respective joint pins and the pivot pin forms an angle  $\alpha$  with the axial direction in the range of from  $30^\circ$  to  $60^\circ$  and in that, in the locking position, the connecting axes between the respective joint pins and the pivot pin form an angle  $\alpha$  with the axial direction in the range of from under  $91^\circ$  to over  $90^\circ$ .

10. Tool changer for a robot, comprising a base unit, a changing unit which can be locked on the base unit, movable locking elements which are provided on the base unit, and receptacles which are provided on the changing unit, the locking elements interacting with the receptacles in a locking position in order to lock the changing unit on the base unit and releasing the receptacles in an unlocking position, said tool changer also comprising a rotary element which is provided on the base unit and is rotatably mounted about its longitudinal axis, the locking elements being movement-coupled to the rotary element in such a way that the locking elements are moved when the rotary element is rotated, characterized in that the rotary element is movement-coupled to a rotationally fixed and axially movable actuation element such that, when the rotary element is rotated, the actuation element changes its position in the axial direction, in that the actuation element is movement-coupled to or forms a pivot pin which extends perpendicularly to the longitudinal axis of the actuation element, in that a joint pin is provided on each of the locking elements, and in that, between the pivot pin and each joint pin, at least one lever arm is provided such that, when the pivot pin is moved in the axial direction, the joint pins and thus also the locking elements are moved in a perpendicular direction running perpendicularly to the axial direction, characterized in that a displaceable axial stop is provided for adjusting the end position of the actuation element in the axial direction.

11. Tool changer for a robot, comprising a base unit, a changing unit which can be locked on the base unit, movable locking elements which are provided on the base unit, and receptacles which are provided on the changing unit, the locking elements interacting with the receptacles in a locking position in order to lock the changing unit on the base unit and releasing the receptacles in an unlocking position, said tool changer also comprising a rotary element which is provided on the base unit and is rotatably mounted about its longitudinal axis, the locking elements being movement-coupled to the rotary element in such a way that the locking elements are moved when the rotary element is rotated, characterized in that the rotary element is movement-coupled to a rotationally fixed and axially movable actuation element such that, when the rotary element is rotated, the actuation element changes its position in the axial direction, in that the actuation element is movement-coupled to or forms a pivot pin which extends perpendicularly to the longitudinal axis of the actuation element, in that a joint pin is provided on each of the locking elements, and in that, between the pivot pin and each joint pin, at least one lever arm is provided such that, when the pivot pin is moved in the axial direction, the joint pins and thus also the locking elements are moved in a perpendicular direction running perpendicularly to the axial direction, characterized in that positioning recesses or positioning rods by means of which the changing unit can be positioned at a changing station are provided on the changing unit.

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